

Homework 1

Note: You may choose to use LaTeX to complete your homework. This will benefit you by familiarizing yourself with the key scientific publishing tool. If you are new to LaTeX, please read [this tutorial](#) first. You may also choose to use pen and paper to complete your homework and scan your solutions.

In this and following assignments, you may need to draw pictures of FA's, PDA's, and TM's in LaTeX. This may be challenging but `tikz` and the `automata` package are the tools that can make things easier. Please find examples at the [manual page](#) to get an idea. On the right (Figure 1), we give an example of a finite automata diagram generated by the automata package.

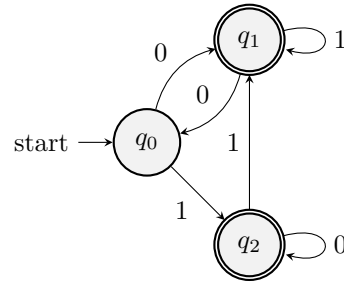


Figure 1: An example of a finite automata.

Problem 1. (Exercise 2.2.2) Let $\hat{\delta}$ be the extended transition function defined in class for DFA. Prove that

$$\hat{\delta}(p, xy) = \hat{\delta}(\hat{\delta}(p, x), y),$$

for all state p and strings x and y .

Solution.

Problem 2. Give DFA's accepting the following languages over the alphabet $\{0, 1\}$:

- (a) (Exercise 2.2.4 (b)) The set of all strings that has three consecutive 0's (not necessarily at the end).
- (b) (Exercise 2.2.4 (c)) The set of strings with 011 as a substring.
- (c) (Exercise 2.2.5 (b)) The set of all strings whose tenth symbol from the right end is a 1.
- (d) (Exercise 2.2.5 (d)) The set of strings such that the number of 0's is divisible by five, and the number of 1's is divisible by 3.

Solution. (a) Write the solution here.

(b)

Problem 3. (Exercise 2.2.9) Let $A = (Q, \Sigma, \delta, q_0, \{q_f\})$ be a DFA, and suppose that for all a in Σ we have $\delta(q_0, a) = \delta(q_f, a)$.

- (a) Show that for all $w \neq \epsilon$, we have $\hat{\delta}(q_0, w) = \hat{\delta}(q_f, w)$.

- (b) Show that if x is a nonempty string in $L(A)$, then for all $k > 0$, x^k (i.e., x written k times) is also in $L(A)$.

Solution.

Problem 4. (Exercise 2.3.2) Convert to a DFA the following NFA:

	0	1
$\rightarrow p$	$\{q, s\}$	$\{q\}$
$*q$	$\{r\}$	$\{q, r\}$
r	$\{s\}$	$\{p\}$
$*s$	$\emptyset$	$\{p\}$

Solution.

Problem 5. (Exercise 2.3.4) Give nondeterministic finite automata to accept the following languages. Try to take advantage of nondeterminism as much as possible.

- The set of strings over alphabet $\{0, 1, \dots, 9\}$ such that the final digit has not appeared before.
- The set of strings of 0's and 1's such that there are two 0's separated by a number of positions that is a multiple of 4. Note that 0 is an allowable multiple of 4.

Solution.